

Neil Malur

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Education

Massachusetts Institute of Technology, BS in Computer Science and Engineering Sept 2021 – May 2025

- GPA: 5.0/5.0 | IEEE Eta Kappa Nu Honor Society | Social and Ethical Responsibilities of Computing Scholar
- **Coursework:** Computer Systems Security, Computer Networks, Natural Language Processing, Software Construction, Computer Vision, Computer Architecture, Neural Circuits for Cognition
- **Activities:** South Asian Association of Students (President), DynaMIT Mentor, AI@MIT Labs, Student Government

Experience

Meta, Software Engineer – New York, NY Sep 2025 – present

- Led development and rollout of safeguards preventing irreversible deletion by AI agents in Meta's core infrastructure, from pretraining workloads to product analytics. Blocked risky endpoints at the service routing layer with safe fallbacks routing to recoverable operations. Protected agent traffic of 1M+ unsafe operations weekly without customer disruption.
- Redesigned core storage client logging for rapid debugging. Introduced tracking for inflight file handles and customized logging enabled per-customer. Additionally, disaggregated latency stats and corrected file size to include unflushed data.
- Reduced critical throttling and latency on hot IO operations. Spearheaded response to failing AI pretraining jobs by increasing data replication of critical files, reducing throttling by over 95%. Collaborated across teams to add a hint boosting hot block replication for our largest customers.
- Led development of code review agent for storage client codebase to accelerate development velocity and defend against harmful code changes. Fine-tuned review criteria and scaled blocking code signals to balance user friction with accuracy.

Nominal, Software Engineer Intern – New York, NY May 2024 – Aug 2024

- Streamlined file ingest by implementing timeseries database read and caching from AWS S3, cutting upload time by 93%. Removed writes to Apache Arrow, instead directing CSV upload to ClickHouse from S3 before caching timeseries data.
- Developed system to generate end-to-end HTTP API connections for user data, built weather data instances for customers. Connections loaded in live and historical data into plotting service optimized for customer ease of use.
- Augmented compute node tree and resolution to support parameterized peak-finding and inverse trigonometric functions
- Investigated thread control changes, including semaphore-enforced limits by user organization and migration to Temporal

United States Department of Commerce, Analyst Intern – Washington, DC May 2023 – Aug 2023

- Developed ML model of domestic macroeconomic zones alongside reusable mapping software to inform federal grants
- Authored economic study of the drivers of Charlotte's transformation with stakeholder interviews and original analysis
- Instituted software pipeline at Commerce, prepared research for Under Secretary and House, and refined bureau strategy

MIT Climate and Sustainability Consortium, Student Researcher – Cambridge, MA May 2022 – Jan 2023

- Created graph learning piece and training strategy of hybrid reinforcement learning system to predict polymer properties.
- Built software interface to simulate crystallization with molecular dynamics, allowing chemists to interact with model
- Coauthored paper presented at ICML Workshop on Generative Modeling

Projects

Security Key Login

- Secured login for a social media website under the WebAuthn 2.0 framework to authenticate second-factor security keys
- Obtained web certificates to communicate via HTTPS by collaborating with an ACME-compliant certificate authority

BERT Prosody Enhancement

- Augmented BERT to accept data with duration-based token embeddings to estimate the impact of textual prosody
- Evaluated model on masked language objective and by correlating attention values, generated data with forced alignment

Technologies

Languages: C++, C, Java, Typescript, Python, Bluespec, React, Node.js, SQL

Software: AWS, PyTorch, TensorFlow, Unity, Docker, ClickHouse, Conjure, Datadog, Temporal